Unit 2 · Session 1 · Tue, Oct 27, 2026

Musculoskeletal Pharmacology — Session 1

4 parts · every fact cited to its slide or textbook page. About 195 minutes of focused study.

Study plan

25 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

100 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

30 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

35 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Tissue injury releases **prostaglandins, bradykinin, histamine, and cytokines** that sensitize nociceptors; **corticosteroids** block phospholipase A_2 upstream, while **NSAIDs** block COX. S6; S9; SG U2-R1; SG U2-R2

2. **Acetaminophen** treats pain and fever but is not a reliable anti-inflammatory; overdose depletes glutathione so **NAPQI** injures the liver, and **N-acetylcysteine** is the antidote. S15; S19; SG U2-R3

3. **COX-1** is the housekeeping enzyme (gastric protection, renal blood flow) and **COX-2** makes inflammatory prostaglandins; **celecoxib** lowers GI risk but raises cardiovascular risk, with similar renal risk. S26; S29; SG U2-R4; SG U2-R5

4. NSAID harms are **GI, renal, and cardiovascular**; flag anticoagulants, ACE inhibitors/ARBs plus diuretics (**triple whammy**), and corticosteroids — and prefer acetaminophen when red flags stack up. S34; S36; S37

5. **OA** is degenerative (treat pain and function); **RA** is autoimmune and needs **DMARDs**, which change the disease course; **methotrexate** is the anchor — **weekly**, with folate, never in pregnancy. S41; S45; S49; SG U2-R6; SG U2-R7

6. **Biologics** target the cascade — TNF (adalimumab, etanercept, infliximab), IL-6 (tocilizumab), B cells (rituximab), T-cell costimulation (abatacept) — and all DMARDs share **infection risk**: screen for TB and hepatitis, vaccinate first. S51; S56; SG U2-R8

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO2.1 Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.

Part 1 · Musculoskeletal Pain & the Inflammatory Cascade
Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.

Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.4 Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-specific factors.

Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.

Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal.

Part 2 · Acetaminophen
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO1 Explain the pharmacologic basis of musculoskeletal pain and inflammation.

Part 1 · Musculoskeletal Pain & the Inflammatory Cascade

SO2 Compare acetaminophen and NSAIDs, including COX-1 vs. COX-2 inhibition.

Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition

SO3 Identify major NSAID adverse effects and select therapy based on patient risk factors.

Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO4 Explain the mechanisms of conventional vs. biologic DMARDs, including representative generic agents in each class.

Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO5 Identify key contraindications, adverse effects, and drug interactions for analgesics and DMARDs.

Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO6 Preview: gout, skeletal muscle relaxants, and bone pharmacology continue in Session 2 (Thursday, October 29).

⚠ Preview only — taught in Session 2 (L04)

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ABOUT THIS LECTURE

- PA Pharmacology Course · Lecture 2 · Session 1 of 2 — Musculoskeletal Pharmacology (Tuesday, October 27): pain, acetaminophen, NSAIDs, and osteoarthritis/rheumatoid arthritis pharmacotherapy, including DMARDs. The session follows one pathway — arachidonic acid — to see where each drug class steps in and how to treat musculoskeletal pain safely. S1; S1 (notes); S2 (notes)
- By the end you should be able to choose between acetaminophen, an NSAID, and a DMARD, and defend the choice. S2 (notes)
- Session 2 (Thursday, October 29) picks up with gout, skeletal muscle relaxants, bone pharmacology, and the cumulative Lecture Recall Quiz. S2 (notes); S61
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah, 2025, McGraw-Hill Education; ISBN 1260463303 eBook; on AccessMedicine: accessmedicine.mhmedical.com/book.aspx?bookid=3382). Unit 2 — Musculoskeletal: Chapters 27, 36, and 42. S3; S3 (notes)
- Recommended text (additional support, not a substitute): Pharmacotherapy Principles and Practice (Chisholm-Burns, Schwinghammer, Malone, Kolesar, Lee, 2022, McGraw-Hill Education; Print ISBN 9781260460278, eText ISBN 9781260460285) — supplemental support alongside Katzung's Chapters 27, 36, and 42 for Unit 2. S3; S3 (notes)
- **Original figures in this deck:** Figures 1–3 (The Arachidonic Acid Pathway: Where Analgesics Act; Nonselective NSAIDs vs. COX-2 Selective Inhibition; Biologic DMARDs: Where Each Class Intercepts the Immune Cascade) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts. The mechanism diagrams added for acetaminophen (including the Rumack-Matthew nomogram schematic), methotrexate and celecoxib, and the NSAID, COX-1 vs. COX-2, conventional DMARD, biologic DMARD and drug-name nomenclature diagrams, are also original illustrative diagrams, created with AI assistance for this course. Companion diagrams for inflammatory mediators, NSAID adverse effects, osteoarthritis vs. rheumatoid arthritis, osteoarthritis pharmacotherapy, disease-modifying drug concepts and DMARD immunosuppression are original illustrative diagrams created with AI assistance for this course as well. They are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S80; S80 (notes)
- **Drug profiles (study reference, S62–S79):** for each drug, formulation, standard adult dose, indications, adverse effects, contraindications and toxicity/overdose — standard adult dosing for teaching; verify against current prescribing information. Abbreviations: PO by mouth; IV intravenous; IM intramuscular; SC subcutaneous; BID/TID twice/three times daily; ER extended

release; DR delayed release; DHFR dihydrofolate reductase; NAPQI the toxic acetaminophen metabolite; NAC N-acetylcysteine. S62; S62 (notes); S63

SLIDE DISCREPANCIES

- **Methotrexate mechanism:** the slides and study guide name dihydrofolate reductase as the target (the depth slide adds a likely adenosine-mediated anti-inflammatory effect at low dose); the textbook says the DHFR effect is relatively minor at RA doses and credits AICAR transformylase, which raises extracellular AMP. Both agree the low-dose effect is anti-inflammatory rather than cytotoxic. Answer course exams with the slide/study-guide version (DHFR). S49; S50; SG U2-M; p. 688
- **Vaccines before DMARDs — two slides differ:** S56 says update vaccines, especially live vaccines, before starting; S57 says update non-live vaccines. Read together: bring all vaccines up to date before therapy, because live vaccines should not be given once immunosuppression has started (textbook: avoid live vaccines up to 3 months after stopping abatacept). S56; S57; S57 (notes); p. 682

1 PART 1 · TOPIC

Musculoskeletal Pain & the Inflammatory Cascade

SO1 IO2.1

Key definitions

- **Nociceptive pain:** direct activation of pain receptors by tissue injury. S5
- **Inflammatory pain:** pain driven by inflammatory mediators. S5
- **Neuropathic pain:** pain from nerve injury or dysfunction. S5
- **Acute pain:** recent onset, often tied to an identifiable injury or flare. S5
- **Chronic pain:** persists beyond expected healing time, often > 3 months. Treatment approach differs sharply between acute and chronic pain. S5

How it works

- **Tissue injury → release of inflammatory mediators → sensitization of nociceptors → pain.** S6
- Key mediators: **prostaglandins, bradykinin, histamine, and cytokines (TNF-α, IL-1, IL-6).** S6; SG U2-R1
- Arachidonic acid → COX enzyme → prostaglandins → pain and inflammation: the spine of the acetaminophen, NSAID, and corticosteroid sections. S8; S8 (notes)
- **NSAIDs and COX-2 inhibitors act at the COX step; glucocorticoids act further upstream; cytokine inhibitors and analgesics act through separate, parallel pathways.** S8
- **Corticosteroids block phospholipase A_2** (membrane phospholipids → arachidonic acid), so a broad range of mediators falls. NSAIDs block only COX, so they lower prostaglandins specifically. S9; SG U2-R2
- **Nociceptor sensitization: injury releases mediators:** PGE2, bradykinin, histamine, and cytokines, which **lower the pain threshold** — normally harmless stimuli now hurt. S7; S6 (notes)
- Prostaglandins sensitize nociceptors, so they are the main NSAID target; bradykinin, histamine, and cytokines are not blocked by NSAIDs. S7 (notes); S12-13
- **Where drugs act:** NSAIDs block COX; corticosteroids block PLA2 (phospholipase A_2) one step upstream. S7; S8 (notes); S9 (notes)
- TB **The immune response:** chronic inflammation involves eicosanoids,

leukotrienes, cytokines, chemokines, and many immune cells; RA, SLE, vasculitis, and gout all stem from this cascade. p. 675

- **TB** The lipoxygenase branch of arachidonic acid makes leukotrienes (chemotaxis, bronchoconstriction, vascular permeability) — a separate branch from the COX step NSAIDs block. p. 675; p. 678

Drugs named as examples

- Corticosteroids (block phospholipase A_2) vs. NSAIDs and COX-2 inhibitors (block COX): two distinct points of intervention. S9; S9 (notes)

CLINICAL

- Most musculoskeletal pain in this lecture is a mix of nociceptive and inflammatory; pure neuropathic pain belongs to a separate pharmacology unit. S5 (notes)
- The mediator list is the roadmap for the lecture: NSAIDs target prostaglandins, biologic DMARDs target specific cytokines. S6 (notes)
- **TB** **Therapeutic strategies:** goal 1 is symptom relief and function (NSAIDs and DMARDs); goal 2 is slowing tissue damage, for which DMARDs are absolutely necessary. Response is tracked with indices such as DAS28 in RA. p. 675

TRAP

- Corticosteroids act **upstream** at phospholipase A_2, not at COX — that is why their effect is broader than an NSAID's. S9; SG U2-R2
- The lecture's drugs target nociceptive and inflammatory pain; neuropathic pain is a different unit. S5 (notes)

Step by step

How injury becomes pain S6

Tissue injury → Inflammatory mediators released → Nociceptors sensitized → Pain

The arachidonic acid pathway S8; S9

Membrane phospholipids → Phospholipase A_2 (blocked by corticosteroids) → Arachidonic acid → COX-1 / COX-2 (blocked by NSAIDs) → Prostaglandins → Pain, inflammation, fever

Inflammatory mediators sensitize nociceptors. Companion diagram: tissue injury → PGE2, bradykinin, histamine, cytokines → nociceptor sensitization → pain. NSAIDs act by blocking COX; corticosteroids by blocking PLA2. S7; S7 (notes)

Figure 1 · The arachidonic acid pathway: where analgesics act. Original illustrative diagram: membrane phospholipids → arachidonic acid → COX-1/COX-2 → prostaglandins → pain, inflammation, fever. Corticosteroids block phospholipase A_2 upstream; NSAIDs and COX-2 inhibitors block the COX step one step later. S9; S9 (notes)

Where each drug class intervenes

Drug class	Step it blocks	What falls
Corticosteroids	**Phospholipase A_2** (upstream)	A broad range of mediators
NSAIDs, COX-2 inhibitors	**COX** enzyme	Prostaglandins specifically
Biologic DMARDs (cytokine inhibitors)	Specific cytokines or immune cells	TNF-α, IL-6 signaling; B or T cells

How to tell them apart. Sources: S8; S9; S6 (notes); SG U2-R2

WHY IT HAPPENS

- Why are corticosteroids broader than NSAIDs? They act at phospholipase A_2, before arachidonic acid is released, so every mediator made from it falls — NSAIDs only stop the COX branch. S9; SG U2-R2

MNEMONIC (MEMORY AID)

- Mediators that make pain: **P-B-H-C** — **P**rostaglandins, **B**radykinin, **H**istamine, **C**ytokines (TNF-α, IL-1, IL-6). S6; SG U2-R1

HIGH-YIELD

- Injury → mediators → nociceptor sensitization → pain. S6
- Musculoskeletal pain is driven by inflammatory mediators (prostaglandins, bradykinin, histamine, cytokines) — NSAIDs and corticosteroids intervene at different points along this cascade. S61; S6; SG U2-R1
- Corticosteroids = phospholipase A_2 (broad); NSAIDs = COX (prostaglandins). S9; SG U2-R2
- Prostaglandins drive inflammatory pain, and COX makes them. S61 (notes)

2 PART 2 · DRUG CLASS

Acetaminophen

SO2 SO5 IO2.1 IO2.2 IO2.6

Class

- Effective analgesia, with a narrow margin in overdose: treats pain and fever well, but is **not an anti-inflammatory drug** the way NSAIDs are. S14; S15 (notes)
- Acetaminophen leaves inflammation alone; NSAIDs go after it directly. S22 (notes)

Prototype + other drugs named

- Acetaminophen (the lecture's single non-NSAID analgesic). S15

Mechanism of action

- **Analgesic effect:** relieves mild-to-moderate pain, with a central mechanism that is still incompletely understood. S15
- **Antipyretic effect:** reduces fever via central effects on the hypothalamic temperature-regulating center. S15
- **Anti-inflammatory effect:** minimal at the periphery — unlike NSAIDs, acetaminophen is not a reliable anti-inflammatory agent. S15
- **Mechanism:** analgesia and antipyresis are mainly central, via reduced CNS prostaglandin synthesis; peripheral anti-inflammatory action is minimal. S16; S15 (notes)
- TB Textbook adds: a weak peripheral COX-1/COX-2 inhibitor whose pain relief also involves opioid, cannabinoid, and serotonergic systems. p. 694

Indications

- Reasonable first-line for mild osteoarthritis pain. S43
- Generally safer first-line than an NSAID in a patient with NSAID red flags (kidney disease, anticoagulation, older age). S37
- TB Preferred over aspirin in aspirin allergy, hemophilia, peptic ulcer history, or aspirin-induced bronchospasm; alone it is inadequate for inflammatory conditions such as RA. p. 694

Adverse effects (boxed warnings first)

- **Overdose: glutathione stores are depleted, NAPQI accumulates, and hepatocyte injury follows.** S19

Contraindications

- **Pearl:** chronic alcohol use and fasting lower GSH and raise risk; both also induce CYP2E1. S17; S17 (notes)

- **Overdose cascade:** glucuronidation/sulfation saturate → more drug shunted to CYP2E1 → more NAPQI, glutathione exhausted → NAPQI binds hepatocyte proteins → **centrilobular liver necrosis.** S17; S17 (notes)
- **TB** Ingesting 15 g may be fatal (centrilobular hepatic necrosis); take care with liver disease, heavy alcohol use, or malnutrition. p. 694

Drug interactions

- **Hidden overdose:** many combination cold, flu, and sleep products also contain acetaminophen — **unintentional double-dosing is a leading cause of acetaminophen-related liver injury.** S20
- **TB** Combining acetaminophen with an NSAID improves pain control but can potentiate the NSAID's GI toxicity. p. 694

Pharmacokinetics

- Normal handling: acetaminophen → NAPQI (toxic metabolite) → glutathione conjugation → safe excretion. S19
- NAPQI = N-acetyl-p-benzoquinone imine — the "toxic metabolite" example from Lecture 1's drug-metabolism section. S19 (notes)
- **Major route (~90%):** glucuronidation and sulfation → safe urinary excretion. S16; S16 (notes)
- **Minor route (~5–10%):** CYP2E1 oxidation → NAPQI (N-acetyl-p-benzoquinone-imine), normally neutralized by glutathione (GSH) conjugation at therapeutic doses. S16; S16 (notes)
- **TB** Mostly conjugated to inactive sulfate and glucuronide; half-life 2–3 hours, relatively unaffected by renal function. p. 694

Monitoring

- **Always ask about ALL sources of acetaminophen.** S20
- Patients often don't realize a product contains acetaminophen unless they check the label — reinforce generic-name awareness. S20 (notes)
- **Acute single ingestion:** plot the acetaminophen level on the **Rumack-Matthew nomogram**; check AST/ALT. S17; S17 (notes)
- **How to use the nomogram:** draw a level at 4 hours or later after a single acute

ingestion and plot it against time since ingestion. S18

- **Above the treatment line:** give NAC. **Below the line:** toxicity unlikely. Example: 8 h, 120 mcg/mL is above the line, so start NAC. S18
- The illustrative treatment line runs from 150 mcg/mL at 4 h, halving every 4 h, to 4.7 mcg/mL at 24 h. S18
- **Limits:** not valid before 4 hours, with an unknown ingestion time, or for staggered, repeated supratherapeutic, or chronic ingestion — use clinical judgment, AST/ALT, and poison control. S18; S18 (notes)
- Start NAC without waiting if presentation is late or the level is unavailable and the ingestion is significant. S18 (notes)
- After 24 hours the nomogram is not used: 4.7 mcg/mL is only where the treatment line ends, not a safe cutoff. A detectable level or elevated AST/ALT guides NAC, per protocol or poison control. S18 (notes)
- TB Adults: generally no more than 4 g/day — and even 4 g is associated with more liver function test abnormalities. p. 694

CLINICAL PEARL / WHY

- **Antidote: N-acetylcysteine**, a glutathione precursor — it replenishes glutathione to detoxify NAPQI. S19; SG U2-R3
- Case: a patient on regular acetaminophen for back pain starts an OTC cold-and-flu combination. **What is the hidden danger in this combination?** Double-dosing acetaminophen. S20
- **Antidote: NAC (N-acetylcysteine) replenishes glutathione; give early for best effect.** S17; S17 (notes)
- Ask: “What is the danger of a cold-and-flu product on top of regular acetaminophen?” Look for unintentional double dosing from combination products. S19 (notes)
- TB **Other analgesics — tramadol:** weak μ-opioid agonist plus serotonin/norepinephrine reuptake inhibition; no anti-inflammatory effect; up to 300 mg/day; risks serotonin syndrome and seizures. p. 694; p. 695

Step by step

Therapeutic dose S16

Oral acetaminophen → liver →

~90%: glucuronidation and sulfation → safe urinary excretion →

~5–10%: CYP2E1 → NAPQI → Glutathione (GSH) conjugation → safe excretion

Overdose and rescue S17; SG U2-R3

Glucuronidation/sulfation saturate → More drug shunted to CYP2E1 →

More NAPQI; glutathione depleted → NAPQI binds hepatocyte proteins →

Centrilobular liver necrosis → NAC replenishes GSH — give early

Using the Rumack-Matthew nomogram S18; S18 (notes)

Single acute ingestion, known time → Level drawn at ≥ 4 h →

Plot level vs. hours since ingestion → Above the treatment line → give NAC →

Below → toxicity unlikely →

Late, chronic, unknown time, or > 24 h → clinical judgment, AST/ALT, poison control

Rumack-Matthew nomogram: assessing toxicity. Illustrative schematic for a single acute ingestion: the treatment line falls from 150 mcg/mL at 4 h to 4.7 mcg/mL at 24 h (halving every 4 h). Above the line: give NAC; below: toxicity unlikely; beyond 24 h the nomogram is not used. Example: 8 h, 120 mcg/mL is above the line. S18; S18 (notes)

Figure 19 · Acetaminophen: safe vs. toxic metabolism. Schematic (illustrative): at therapeutic doses ~90% of acetaminophen is glucuronidated or sulfated and excreted safely; the minor CYP2E1 route (~5–10%) makes NAPQI, which glutathione neutralizes. S16; S16 (notes)

Figure 4 · Acetaminophen overdose and NAC rescue. Schematic (illustrative): overdose saturates glucuronidation and sulfation, shunting drug to CYP2E1; NAPQI accumulates once glutathione is exhausted, binds hepatocyte proteins and causes centrilobular necrosis. NAC replenishes glutathione. S17; S17 (notes)

Acetaminophen vs. NSAIDs

	Acetaminophen	NSAIDs
Pain	Yes (mild–moderate)	Yes
Fever	Yes (hypothalamus)	Yes
Inflammation	**Minimal — not reliable**	**Yes — powerful**
Main danger	Liver (NAPQI) in overdose or double-dosing	GI, renal, cardiovascular
Antidote	N-acetylcysteine	Not covered in materials

How to tell them apart. Sources: S15; S19; S23; S34

Acetaminophen metabolism: therapeutic dose vs. overdose

	Therapeutic dose	Overdose
Glucuronidation + sulfation	~90% → safe urinary excretion	**Saturated**
CYP2E1 → NAPQI	~5–10%	**More drug shunted here**
Glutathione (GSH)	Neutralizes NAPQI	**Exhausted** → NAPQI binds hepatocytes

	Therapeutic dose	Overdose
Result	Safe excretion	Centrilobular liver necrosis — give NAC early

How to tell them apart. Sources: S16; S17; SG U2-R3

Acetaminophen drug profile: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Acetaminophen (Tylenol)	**Analgesic / antipyretic**	Tablets 325/500 mg; ER 650 mg; liquid; suppository; IV 1,000 mg	650-1,000 mg PO every 6 hours (maximum 3-4 g/day; 2-3 g/day with chronic alcohol use or liver disease)	Mild to moderate pain, fever, osteoarthritis pain

How to tell them apart. Sources: S63

Acetaminophen drug profile: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Acetaminophen (Tylenol)	Rare at therapeutic doses; rash; hepatotoxicity with excess	Severe hepatic impairment; caution with chronic alcohol use	Hepatotoxicity from NAPQI after overdose; treat with N-acetylcysteine, ideally within 8 hours; use the Rumack-Matthew nomogram

How to tell them apart. Sources: S63

WHY IT HAPPENS

- Why does N-acetylcysteine work? Glutathione is what safely conjugates NAPQI. Overdose uses it up; N-acetylcysteine is a glutathione precursor, so it refills the supply. S19; SG U2-R3

MNEMONIC (MEMORY AID)

- Acetaminophen does **P**ain and **P**yrexia, not **P**uffiness (inflammation). S15

HIGH-YIELD

- Acetaminophen treats pain and fever but is not a reliable anti-inflammatory agent; its overdose toxicity is mediated by NAPQI accumulation. S61; S15; S19
- Overdose: glutathione depleted → NAPQI → hepatocyte injury. S19
- Antidote: N-acetylcysteine (glutathione precursor). S19; SG U2-R3
- Ask about ALL sources — combination cold/flu/sleep products hide acetaminophen. S20
- Acetaminophen is central, with no real anti-inflammatory effect. Overdose means NAPQI, and the antidote is NAC. S61 (notes)

3 PART 3 · DRUG CLASS

NSAIDs & COX-2 Selective Inhibition

SO2 SO3 SO5 IO2.1 IO2.2 IO2.3 IO2.4

Class

- Powerful anti-inflammatory effect, with real systemic risk. S23
- **Nonselective NSAIDs inhibit both COX-1 and COX-2.** S28
- **Celecoxib selectively inhibits COX-2 over COX-1.** S29

Prototype + other drugs named

- Ibuprofen — common first-line choice for mild-moderate pain. S28
- Naproxen — longer half-life, twice-daily dosing. S28
- Diclofenac — available in oral and topical formulations. S28
- Indomethacin — more potent, more GI/CNS adverse effects. S28
- **Ketorolac — short-term use only, significant GI/renal risk.** S28
- Celecoxib — the COX-2 selective agent. S29

Mechanism of action

- NSAID → COX inhibition → ↓ prostaglandins → ↓ pain, inflammation, fever. S24
- **COX-1, the "housekeeping" enzyme:** maintains gastric mucosal protection and normal renal blood flow; inhibiting it causes GI and renal adverse effects. S26; SG U2-R4
- **COX-2:** induced mainly at sites of inflammation and the primary source of inflammatory prostaglandins; selective inhibition targets inflammation with less GI toxicity. S26; SG U2-R4
- Simplified teaching model, not an absolute separation: COX-1 and COX-2 functions overlap, and COX-2 is also constitutively expressed in the kidney and elsewhere. S26 (notes)
- **Pathway:** COX converts arachidonic acid to prostaglandins **and thromboxane A2 (TXA2)**. Less prostaglandin: pain, fever,

Indications

- Oral NSAIDs for osteoarthritis — effective, but carry the systemic risks below. S43
- Topical NSAIDs — effective local relief with much lower systemic exposure. S43
- TB NSAIDs ease symptoms but, unlike DMARDs, do not alter the course of any arthritic disorder. p. 676
- TB Used in RA, spondyloarthropathies, OA, sprains and strains, low back pain, and gout (tolmetin appears ineffective in gout). p. 676

and inflammation fall. Less TXA2: bleeding risk. S25; S24 (notes)

- Every NSAID shares the COX mechanism; the side effects follow from losing protective prostaglandins (GI injury, platelet inhibition, renal effects). S25; S25 (notes)
- **COX-1 is constitutive:** gastric mucosa (protection), platelets (TXA2, clotting), kidney (renal blood flow). **COX-2 is inducible** in inflamed tissue: pain, swelling, fever. S27; S26 (notes)
- Nonselective NSAIDs block both isoforms, so GI injury is common. Selectivity trades anti-inflammatory benefit against GI and platelet protection. S27; S27 (notes)
- Celecoxib selectively inhibits inducible COX-2, the main source of inflammatory prostaglandins. **GI benefit:** stomach COX-1 is spared, lowering ulcer and bleeding risk. S30; S27
- **Celecoxib's trade-off:** PGI2 (prostacyclin) falls without a matching drop in TXA2, which raises cardiovascular event risk; stomach and platelet protection continue. S30; S29 (notes)
- Celecoxib still affects kidney prostaglandins, so renal risk is not eliminated. S30; S29 (notes)
- **TB** Aspirin irreversibly blocks platelet COX (the effect lasts the 8–10-day life of the platelet); other nonselective NSAIDs are reversible inhibitors. p. 675; p. 677
- **TB** COX-2 inhibitors spare platelet thromboxane (made by COX-1) but block endothelial prostacyclin — so they are not cardioprotective. p. 677

Adverse effects (boxed warnings first)

- **Gastrointestinal:** gastritis, peptic ulcer, GI bleeding. S34
- **Renal:** decreased renal perfusion, fluid retention, acute kidney injury. S34
- **Cardiovascular:** hypertension, increased risk of cardiovascular events. S34

Contraindications

- Use caution or avoid in: active GI bleeding or ulcer history; chronic kidney disease; heart failure or cardiovascular disease. S36
- Also caution or avoid with anticoagulant therapy and in late pregnancy. S36
- **TB** Ibuprofen is relatively contraindicated with nasal polyps, angioedema, or bronchospastic reactivity to aspirin. p. 680

- **Celecoxib:** lower risk of ulcer/bleeding, but **COX-2 selectivity is associated with increased cardiovascular event risk — a key trade-off.** S29; SG U2-R5
- Celecoxib's renal risk is similar to nonselective NSAIDs — COX-2 selectivity does not protect the kidney. S29; SG U2-R5
- **GI:** less prostaglandin-driven mucus and bicarbonate leaves the mucosa open to injury (ulcer, bleeding). S35; S34 (notes)
- **Kidney:** reduced afferent arteriolar dilation lowers renal blood flow (AKI, fluid retention). S35; S34 (notes)
- **Platelets:** reduced TXA2 impairs aggregation, increasing bleeding risk. S35
- **TB** Beyond GI, renal, and CV: headache, tinnitus, dizziness, rare aseptic meningitis, liver test abnormalities, asthma, rashes, hyperkalemia. p. 676

Drug interactions

- NSAIDs + anticoagulants → ↑ bleeding risk. S36
- NSAIDs + ACE inhibitors/ARBs → ↑ AKI risk; add a diuretic and it is the **"triple whammy,"** a very high-yield AKI pattern. S36; S36 (notes)
- NSAIDs + diuretics → reduced diuretic effect, ↑ renal risk. S36
- NSAIDs + corticosteroids → ↑ GI ulcer risk. S36
- **TB** Ibuprofen taken with aspirin blocks aspirin's irreversible platelet inhibition, limiting its cardioprotective effect. p. 680
- **TB** Celecoxib (a CYP2C9 substrate) occasionally interacts with warfarin and raises lithium and fluconazole levels. p. 679

Pharmacokinetics

- Naproxen's longer half-life allows twice-daily dosing; diclofenac comes as oral and topical forms. S28
- **TB** **Chemistry & pharmacokinetics:** most NSAIDs are weak acids, well absorbed, ~98% albumin-bound, metabolized by CYP3A or CYP2C, and renally excreted. p. 675

Monitoring

- These three systems (GI, renal, CV) are the organizing framework for every NSAID contraindication and interaction discussion (Table 1). S34 (notes)
- **TB** Highest GI-bleeding risk: celecoxib, or a nonselective NSAID plus omeprazole

or misoprostol. p. 681

CLINICAL PEARL / WHY

- Case: a 72-year-old with OA, CKD, hypertension, and an anticoagulant. **Is an NSAID a good first-line choice for this patient's pain? Why or why not?** S37
- **Three NSAID red flags at once** — renal disease, anticoagulation, and age-related bleeding risk — make acetaminophen or a non-NSAID strategy generally safer first-line. S37
- Celecoxib is a classic "trade one risk for another" point: a better GI profile, but not a free lunch cardiovascularly or renally. S29 (notes)
- There is no risk-free NSAID choice, only a different risk profile. S31 (notes)
- TB **Choice of NSAID:** all are about equally effective, so choose by toxicity, cost, and patient factors; in renal insufficiency nonacetylated salicylates may be best. p. 681

Step by step

How NSAIDs work S24; S25

NSAID → COX inhibition → ↓ Prostaglandins and TXA2 →

↓ Pain, inflammation, fever (and ↑ bleeding risk)

Why celecoxib raises cardiovascular risk S29 (notes); S30

Celecoxib blocks COX-2 → Endothelial PGI2 falls →

Platelet TXA2 (COX-1) preserved → Balance tips toward clotting →

↑ Cardiovascular events

Is an NSAID the right first choice? S34; S36; S37; S43

Check GI: ulcer or bleeding history? → Check kidney: CKD, ACEi/ARB, diuretic?

→ Check heart: HF, CV disease, hypertension? →

Check bleeding: anticoagulant, older age? →

Red flags → acetaminophen or topical/non-NSAID strategy

Figure 2 · Nonselective NSAIDs vs. COX-2 selective inhibition. Original illustrative comparison (illustrative relative-risk values, not from Katzung; not clinical data): nonselective NSAIDs (e.g., ibuprofen, naproxen) carry higher GI ulcer/bleeding risk (75) than cardiovascular risk (35); celecoxib trades lower GI risk (30) for higher cardiovascular risk (65). S31; S31 (notes)

NSAID GI, renal and platelet effects. Companion diagram: reduced prostaglandins → GI mucosal injury (less mucus and bicarbonate, ulcer risk), reduced afferent arteriolar dilation (kidney blood flow falls), and reduced TXA2 (platelets aggregate less, bleeding risk). S35; S34 (notes)

Figure 6 · NSAIDs block COX and prostaglandin synthesis. Schematic (illustrative): membrane phospholipids → arachidonic acid → COX-1/COX-2 → prostaglandins and TXA2. NSAIDs block COX, so pain, fever and inflammation fall — and so does TXA2 (bleeding risk). S25; S25 (notes)

Figure 7 · COX-1 vs. COX-2: constitutive vs. inducible. Schematic (illustrative): COX-1 is constitutive (stomach mucosal protection, platelet TXA2, renal blood flow); COX-2 is induced in inflamed tissue (pain, swelling, fever). Nonselective NSAIDs block both, so GI injury is common. S27; S27 (notes)

Figure 8 · Celecoxib: sparing COX-1 while blocking COX-2. Schematic (illustrative): celecoxib blocks COX-2 (fewer inflammatory prostaglandins) and spares COX-1 (stomach and platelet protection continue); PGI2 falls while TXA2 is kept — the cardiovascular trade-off. S30; S30 (notes)

COX-1 vs. COX-2

	COX-1	COX-2
Expression	**Constitutive** ("housekeeping")	**Inducible** in inflamed tissue
Protects / makes	Gastric mucosa; platelet TXA_2 (clotting); renal blood flow	Primary source of **inflammatory prostaglandins**
Blocking it causes	GI and renal adverse effects; less clotting	Less pain, swelling, fever (selective blockade = less GI toxicity)
Who blocks it	Nonselective NSAIDs	Nonselective NSAIDs and celecoxib

How to tell them apart. Sources: S26; S27; SG U2-R4

Nonselective NSAIDs vs. celecoxib

	Nonselective (ibuprofen, naproxen…)	Celecoxib (COX-2 selective)
GI ulcer / bleeding	**Higher** (Figure 2: 75)	**Lower** (Figure 2: 30)
Cardiovascular events	Lower (Figure 2: 35)	**Higher** (Figure 2: 65)

	Nonselective (ibuprofen, naproxen…)	Celecoxib (COX-2 selective)
Renal risk	Present	**Similar** — selectivity does not protect the kidney

How to tell them apart. Sources: S29; S31; SG U2-R5

NSAID drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Ibuprofen (Advil, Motrin)	**Nonselective NSAID**	Tablets 200/400/600/800 mg; suspension; IV	200-800 mg PO every 6-8 hours with food (maximum 3,200 mg/day Rx; 1,200 mg OTC)	Pain, inflammation, fever, osteoarthritis, rheumatoid arthritis, dysmenorrhea
Naproxen (Naprosyn, Aleve)	**Nonselective NSAID**	Tablets 220/250/375/500 mg; DR and ER forms; suspension	250-500 mg PO twice daily with food (maximum 1,250 mg/day)	Osteoarthritis, rheumatoid arthritis, gout flare, tendinitis, dysmenorrhea
Diclofenac (Voltaren, Cataflam)	**Nonselective NSAID (COX-2 preferring)**	Tablets 50 mg; DR 75 mg; ER 100 mg; 1% gel; ophthalmic; patch	50 mg PO two to three times daily or 75 mg twice daily (maximum 150 mg/day); gel 2-4 g to joint up to four times daily	Osteoarthritis, rheumatoid arthritis, acute pain, topical for knee and hand osteoarthritis
Ketorolac (Toradol)	**Nonselective NSAID (short-term, potent)**	Tablets 10 mg; injection 15/30 mg; nasal spray	15-30 mg IV/IM every 6 hours (maximum 120 mg/day) or 10 mg PO every 4-6 hours; total course 5 days or less	Short-term management of moderately severe acute pain
Celecoxib (Celebrex)	**Selective COX-2 inhibitor**	Capsules 50/100/200/400 mg	200 mg PO once daily or 100 mg twice daily (rheumatoid arthritis 100-200 mg twice daily)	Osteoarthritis, rheumatoid arthritis, acute pain, ankylosing spondylitis

How to tell them apart. Sources: S64; S65; S66; S67; S68

NSAID drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Ibuprofen (Advil, Motrin)	GI upset, ulcer and bleeding, renal injury, hypertension, fluid retention, platelet inhibition	Active GI bleed, NSAID-triggered asthma, CABG setting, advanced renal disease, third trimester pregnancy, heart failure (caution)	GI bleeding, acute kidney injury, metabolic acidosis, seizures in large overdose; supportive care

Drug	Adverse effects	Contraindications	Toxicity / overdose
Naproxen (Naprosyn, Aleve)	GI upset and bleeding, renal injury, hypertension, fluid retention, cardiovascular risk	Active GI bleed, NSAID-triggered asthma, CABG setting, severe renal impairment, third trimester	GI bleeding, renal injury, lethargy and seizures in overdose; activated charcoal and supportive care
Diclofenac (Voltaren, Cataflam)	GI upset and bleeding, elevated liver enzymes, hypertension, renal injury, cardiovascular risk	Active GI bleed, CABG setting, severe renal or hepatic impairment, NSAID-triggered asthma	Hepatotoxicity, GI bleeding, renal injury; topical form has much lower systemic exposure
Ketorolac (Toradol)	GI bleeding, renal injury, bleeding, hypertension	Active GI bleed, renal impairment, high bleeding risk, perioperative CABG, labor and delivery, pregnancy, age 65+ (reduce dose)	Boxed warnings for GI bleeding, renal failure, bleeding and cardiovascular events; limit to 5 days
Celecoxib (Celebrex)	Dyspepsia, hypertension, edema, renal injury, cardiovascular thrombotic risk	Sulfonamide allergy, CABG setting, active GI bleed, advanced renal disease, aspirin-triggered asthma (caution)	Boxed warning for cardiovascular and GI risk; renal injury in overdose; supportive care

How to tell them apart. Sources: S64; S65; S66; S67; S68

Other NSAIDs in the reading (textbook)

Drug	Half-life (h)	What sets it apart
Aspirin	0.25	Now mainly antiplatelet (81–325 mg daily); avoid in hemophilia
Nonacetylated salicylates	Not covered in materials	Anti-inflammatory without platelet inhibition: an option in asthma, bleeding tendency, renal dysfunction
Ibuprofen	2	≤ 1600 mg/day analgesic only; 2400 mg/day anti-inflammatory
Naproxen	12–17	More GI bleeding than OTC ibuprofen, better CV profile; avoid in hepatic impairment
Diclofenac	1.9–2.3	More aminotransferase rises; 1% topical gel OTC
Indomethacin	4–5	Headache in 11.7%; can worsen depression, epilepsy, parkinsonism
Ketorolac	5–6	Analgesic use; start IV/IM/intranasal; ≤ 120 mg/day; ≤ 5 days
Celecoxib	11.2	10–20× COX-2 selective; 100–200 mg bid; no platelet effect at usual doses
Meloxicam	20	Relatively COX-2 selective at 7.5 mg/day; fewer GI effects
Etodolac	6.4	Relatively COX-2 selective; linked to acute pancreatitis

Drug	Half-life (h)	What sets it apart
Nabumetone	22.5–29.8	Only nonacid NSAID (prodrug); less ulceration
Piroxicam	50	Above 20 mg/day sharply raises ulcer and bleeding risk
Sulindac	16.4	Less effect on renal prostaglandins; more liver test abnormalities
Oxaprozin · diflunisal · flurbiprofen · tolmetin	54.9 · 8–12 · 3.8 · 1–2	Oxaprozin once daily; diflunisal rarely used; flurbiprofen rare tremor/rigidity; tolmetin ineffective in gout

How to tell them apart. Sources: p. 677; p. 679; p. 680; p. 681; p. 694

WHY IT HAPPENS

- Why do NSAIDs hurt the stomach and kidney? Blocking COX-1, the housekeeping enzyme, removes the prostaglandins that protect gastric mucosa and keep renal blood flow normal. S26
- Why the “triple whammy”? An ACE inhibitor/ARB, a diuretic, and an NSAID each push kidney perfusion the wrong way at once — together they cause acute kidney injury. S36; S36 (notes)

MNEMONIC (MEMORY AID)

- NSAID harms = **GRC**: **G**ut, **R**enal, **C**ardiovascular (Table 1). S34

HIGH-YIELD

- COX-1 = housekeeping (gastric protection, renal blood flow); COX-2 = inflammation. S26; SG U2-R4
- NSAIDs are effective but carry real GI, renal, and cardiovascular risk; celecoxib trades GI risk for cardiovascular risk (renal risk similar). S61; S29; SG U2-R5
- NSAID harms: GI, renal, cardiovascular. S34
- Triple whammy: NSAID + ACEi/ARB + diuretic → AKI. S36 (notes)
- Ketorolac: short-term only; significant GI/renal risk. S28
- COX-1 protects the stomach, platelets and kidney. Nonselective NSAIDs block both isoforms. S61 (notes)
- Celecoxib cuts GI risk but raises cardiovascular risk, and does not spare the kidney. S61 (notes)

4 PART 4 · DRUG CLASS

Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.1 IO2.2 IO2.3 IO2.4 IO2.5 IO2.6 SO3 SO4 SO5

Class

- **Osteoarthritis (OA):** primarily degenerative joint disease — cartilage breakdown with mechanical wear. S41
- **Rheumatoid arthritis (RA):** autoimmune, systemic inflammatory disease — immune-mediated joint destruction. S41
- **OA pathophysiology:** mechanical load and wear degrade cartilage; bone remodels with osteophytes. S42; S41 (notes)
- **RA pathophysiology:** immune activation → autoimmune synovitis forms a **pannus** that erodes cartilage and bone; **TNF-α and IL-6** drive the inflammation and joint damage. S42; S41 (notes)
- OA treatment goal: pain control and function. S41
- RA treatment goal: suppress disease activity, not just pain. S41
- The hinge: OA is managed mainly with analgesics/NSAIDs, while RA requires disease-modifying therapy. S41 (notes)
- **Key RA cytokines:** TNF-α and IL-6 drive inflammation and joint damage — the targets of the biologic DMARDs. S42
- **DMARD (Disease-Modifying Antirheumatic Drug):** a drug intended to alter the course of RA, not just mask its symptoms. S45; SG U2-R6
- **Goal of therapy:** slow or halt joint destruction and systemic inflammation, rather than providing analgesia alone. S45
- An NSAID can make a patient feel better without changing the disease trajectory; a DMARD is meant to change the trajectory itself. S45 (notes); SG U2-R6

Prototype + other drugs named

- **Methotrexate — the anchor drug for most RA regimens (next slide).** S47
- Hydroxychloroquine — milder disease; relatively favorable safety profile. S47
- Sulfasalazine — often used in combination regimens. S47
- Leflunomide — alternative when methotrexate is not tolerated. S47
- Adalimumab, etanercept, infliximab — **TNF inhibitors**. S51; SG U2-R8
- Tocilizumab — **IL-6 inhibitor**. S51; SG U2-R8
- Rituximab — **B-cell targeted therapy**. S51; SG U2-R8
- Abatacept — **T-cell costimulation modulator**. S51; SG U2-R8

- **NSAIDs** relieve pain and swelling within days but do not alter joint damage; **DMARDs** take weeks to months to act, then slow cartilage and bone destruction. S46; S46 (notes)
- **Timing:** judge a DMARD over months, not after the first visit — set expectations that benefit builds over months. Relief now with NSAIDs, protection later with DMARDs. S46; S46 (notes)
- Targeted therapy: each biologic class corresponds to a different cytokine or cell type in the inflammatory cascade — the mediator list from the start of the lecture. S51 (notes)
- TB DMARD effects take 2 weeks to 6 months to appear. Types: conventional synthetic (cs), biologic (b), and targeted synthetic (ts, e.g., tofacitinib). p. 681
- TB Biologics are large molecules, usually proteins made by recombinant DNA technology; JAK inhibitors are oral small-molecule targeted synthetic DMARDs. p. 681; p. 686

Mechanism of action

- **Mechanism: inhibits dihydrofolate reductase, reducing immune cell proliferation at low RA doses.** (methotrexate) S49; SG U2-M
- **Hydroxychloroquine** alters lysosomal and TLR signaling (less antigen signaling); **sulfasalazine** reduces cytokine and prostaglandin production. S48
- **Leflunomide** inhibits DHODH, blocking pyrimidine synthesis (slower T-cell growth). S48; S47 (notes)
- Conventional DMARDs act inside cells or on broad immune pathways, with a slower onset than biologics. S48 (notes)
- **Methotrexate's folate pathway:** blocking DHFR stops dihydrofolate (DHF) → tetrahydrofolate (THF), the folate needed for purine and thymidylate (DNA) synthesis. S50; S50 (notes)
- **Low-dose RA effect:** methotrexate is likely anti-inflammatory via **adenosine release**,

Indications

- **OA step 1:** acetaminophen — reasonable first-line for mild OA pain. S43
- **OA step 2:** oral NSAIDs (effective, systemic risks) or topical NSAIDs (local relief, much lower systemic exposure). S43
- **Intra-articular corticosteroid injections:** short-term relief for flares in a single joint. S43
- Other approaches: activity modification, physical therapy, weight management. S43
- **Where each OA analgesic acts:** acetaminophen centrally (brain and spinal cord COX) — little anti-inflammatory effect in the joint. S44; S44 (notes)
- **Oral or topical NSAIDs:** block COX in joint tissue (synovium and cartilage), lowering prostaglandins, pain, and swelling. S44; S44 (notes)
- **Intra-articular steroid:** calms local inflammation inside the joint for fast, short-

not cytotoxicity. S50
- TNF inhibitors block tumor necrosis factor-alpha signaling. S51
- IL-6 inhibitors block interleukin-6 receptor signaling. S51
- B-cell targeted therapy depletes or modulates B lymphocytes. S51
- T-cell costimulation modulators block the second signal needed for T-cell activation. S51; SG U2-M
- Figure 3: abatacept acts at T-cell activation, TNF inhibitors and tocilizumab at cytokine release, rituximab at B cells — all upstream of synovial inflammation and joint destruction. S53
- **TNF inhibitors:** antibodies (infliximab, adalimumab) and etanercept, a receptor fusion protein, neutralize soluble and membrane TNF-α, preventing TNFR activation. S52
- **Tocilizumab** blocks both membrane and soluble IL-6 receptors. S52
- **Abatacept (CTLA-4-Ig)** binds CD80/86 and blocks CD28 costimulation, acting early at T-cell activation. S52; S51 (notes)
- **Rituximab (anti-CD20)** binds CD20 and depletes B cells for months. S52; S51 (notes)
- Each biologic intercepts a different step of the immune cascade: walk it from T-cell activation to cytokines to synovial damage — abatacept acts earliest. S52 (notes); S53 (notes)
- **IL-6 blockers:** tocilizumab blocks the IL-6 receptor; it can raise lipids and liver enzymes. S52
- **Abatacept:** blocks CD28 costimulation at T-cell activation. **Rituximab** binds CD20 and depletes B cells. S52

term flare relief. Choice depends on which tissue drives the pain. S44; S44 (notes)
- Methotrexate anchors most RA regimens; leflunomide is the alternative when it is not tolerated. S47
- Rheumatoid arthritis: they modify the disease course rather than only masking symptoms. S61; SG U2-R6
- TNF inhibitors are the first-line biologic; tocilizumab works as monotherapy; rituximab is used after TNF-inhibitor failure. S52
- TB **Glucocorticoid drugs in rheumatic diseases:** used chronically in 30–70% of RA patients; relief is prompt and dramatic, and they slow new bone erosions. p. 693
- TB Long-term prednisone in RA: no more than 7.5 mg daily, tapered when possible — even 2.8 mg daily for over 5 years is linked to severe infections. p. 693
- TB Intra-articular corticosteroids often relieve painful symptoms for up to several months. p. 693
- TB Long-term glucocorticoid harms: osteoporosis, hyperglycemia, Cushing syndrome, infections, cataract, glaucoma, gastric ulcer, depression. p. 694
- TB Methotrexate starts at 7.5 mg weekly and rises to 15–25 mg weekly; it slows new erosions in RA. p. 688
- TB **Combination therapy with DMARDs** is now the rule when monotherapy falls short; triple therapy (methotrexate, sulfasalazine, hydroxychloroquine) rivals etanercept plus methotrexate. p. 693

Adverse effects (boxed warnings first)

- TB **Boxed warning — all TNF inhibitors:** possible malignancy, especially lymphoma. p. 692

Contraindications

- **Methotrexate is contraindicated in pregnancy (teratogenic).** S49; S50 (notes)
- TB Don’t stack biologics: abatacept with a TNF inhibitor probably raises serious infections; upadacitinib with biologic

- TB **Boxed warning — all JAK inhibitors:** venous thromboembolism (DVT, PE), major adverse cardiovascular events, and cancer. p. 686
- Methotrexate key adverse effects: **hepatotoxicity, bone marrow suppression, pulmonary toxicity.** S49; SG U2-R7
- **The major safety theme across all DMARDs: immunosuppression → increased infection risk.** This holds for conventional and biologic DMARDs alike. S56; SG U2-R8
- **Risk:** suppressed immunity lets latent TB and opportunistic infections (TB, viral, fungal) emerge. S57
- TNF inhibitors: watch for reactivation of latent TB. Tocilizumab can raise lipids and liver enzymes. Abatacept has a generally lower infection signal. S52
- TB Methotrexate's most common toxicities are nausea and stomatitis; dose-related liver enzyme elevations are frequent. p. 689
- TB TNF inhibitors: TB reactivation, fungal infections, hepatitis B reactivation, new or worse heart failure, demyelinating disease, lupus-like syndrome. p. 690; p. 692

DMARDs is not recommended. p. 682; p. 687

Drug interactions

- TB Hydroxychloroquine can reduce methotrexate clearance. p. 688
- TB Tocilizumab stimulates CYP450 — doses of cyclosporine or warfarin may need adjusting. p. 692
- TB Methotrexate given with TNF inhibitors reduces anti-drug antibodies (infliximab, golimumab, certolizumab). p. 691; p. 692

Pharmacokinetics

- **Methotrexate is dosed weekly (not daily) — a critical patient-safety point, since daily dosing can be fatal.** S49; SG U2-R7
- **Weekly dosing only:** daily dosing has caused fatal toxicity. S50; S49 (notes)

Monitoring

- **Folate supplementation** is routinely co-prescribed with methotrexate to reduce toxicity. S49; SG U2-R7

- Folic acid co-therapy reduces methotrexate's liver, marrow, lung, and GI toxicity. S50; S50 (notes)
- **Baseline screening:** e.g., tuberculosis and hepatitis B/C screening before starting biologics. S56
- **Vaccination:** update vaccines, especially live vaccines, before starting therapy. S56
- **Ongoing monitoring:** regular labs to catch cytopenias, hepatotoxicity, or infection early. S56
- **Infection precautions:** counsel patients to report fevers or infection symptoms promptly. S56
- **Screen first:** TB test (IGRA or TST) and hepatitis B and C serology before starting; check hepatitis B before rituximab. S57; S52
- **Vaccines:** update non-live vaccines before treatment. S57; S57 (notes)
- **Monitor:** CBC and liver and kidney labs every 1–3 months; act early on fever or infection signs. S57; S57 (notes)
- **TB** Folic acid or leucovorin reduces stomatitis, GI, and liver test problems but may cut methotrexate efficacy by 10–18%. p. 689
- **TB** Hydroxychloroquine: baseline eye exam, then annual screening after 5 years (retinopathy). p. 684
- **TB** Avoid live vaccines for up to 3 months after stopping abatacept; stop upadacitinib before a live attenuated vaccine. p. 682; p. 687

CLINICAL PEARL / WHY

- A 72-year-old with OA, CKD, hypertension, and an anticoagulant: acetaminophen or a non-NSAID strategy first. S37
- OA and RA both affect the joint but are two very different diseases — don't give RA patients analgesia alone. S40; S41 (notes)
- Methotrexate's weekly dosing is one of the highest-yield patient-safety points in all of rheumatology pharmacology. S49 (notes)
- Match each generic name to its class: adalimumab, etanercept, infliximab (TNF); tocilizumab (IL-6); rituximab (B cell); abatacept (T-cell costimulation). S51; S53; S61
- **Read the stem first:** it reveals the mechanism class before the drug is recalled (-mab antibody: -zumab humanized, -ximab chimeric, -umab fully human; -cept receptor fusion; -nib/-tinib kinase

inhibitor; -coxib COX-2 selective; -profen, -fenac, -oxicam NSAID subclasses). S60; S60 (notes)

- TB Topical diclofenac 1% gel is sold over the counter for OA and muscular pain. p. 679
- TB **Biosimilars:** near-copies of legacy biologics with the same efficacy, toxicity, and immunogenicity at potentially lower cost. p. 693
- TB If methotrexate alone falls short: add csDMARDs or a biologic — usually a TNF inhibitor first, then non-TNF biologics or JAK inhibitors. p. 701

Step by step

Methotrexate safety S49; SG U2-R7

Weekly dose (never daily) → Folate co-prescribed →

Watch liver, marrow, lung → Avoid in pregnancy (teratogenic)

Methotrexate and the folate pathway S50

Dihydrofolate (DHF) → DHFR — blocked by methotrexate →

Tetrahydrofolate (THF) → Purine and thymidylate (DNA) synthesis →

Low dose: anti-inflammatory (adenosine release) → Weekly dosing only

Starting a biologic S56

Screen: TB, hepatitis B/C → Update vaccines (especially live) →

Start therapy → Regular labs: cytopenias, liver, infection →

Report fever or infection promptly

Osteoarthritis vs. rheumatoid arthritis: pathophysiology. Companion diagram: OA (mechanical) runs mechanical load and wear → cartilage degradation → bone remodeling and osteophytes; RA (autoimmune) runs immune activation → synovitis with pannus → TNF-α and IL-6 drive joint erosion. S42; S41 (notes)

Where each analgesic acts in osteoarthritis. Companion diagram: acetaminophen (oral) acts centrally on brain and spinal cord COX — analgesia and antipyresis, little joint anti-inflammation; oral or topical NSAIDs block COX in synovium and cartilage — less prostaglandin, pain, and swelling; an intra-articular corticosteroid calms local inflammation for fast, short-term flare relief. S44; S44 (notes)

Symptom relief vs. slowing joint destruction. Schematic, not data: symptom relief from NSAIDs rises within days and plateaus; joint-damage slowing from DMARDs builds over weeks to months. S46; S46 (notes)

Figure 14 · Conventional DMARDs: where each drug acts. Schematic (illustrative): methotrexate inhibits DHFR (slower immune-cell growth); hydroxychloroquine alters lysosome and TLR signals (less antigen signaling); sulfasalazine cuts cytokines and prostaglandins; leflunomide inhibits DHODH (slower T-cell growth). S48; S48 (notes)

Figure 15 · Methotrexate: folate pathway and low-dose action. Schematic (illustrative): methotrexate blocks dihydrofolate reductase (DHFR), so dihydrofolate is not converted to tetrahydrofolate for DNA synthesis; at low RA doses the effect is likely anti-inflammatory (adenosine). Weekly dosing only. S50; S50 (notes)

Figure 3 · Biologic DMARDs: where each class intercepts the immune cascade. Original illustrative diagram (not from Katzung): abatacept blocks T-cell costimulation; TNF inhibitors (adalimumab, etanercept, infliximab) and tocilizumab block cytokine signaling; rituximab depletes B cells — each upstream of synovial inflammation and joint destruction. Walk it from T-cell activation to cytokines to synovial damage: abatacept acts earliest. S53; S53 (notes)

Immune suppression raises infection risk. Companion diagram: a DMARD or biologic suppresses immunity → infection risk (TB, viral, fungal) → screen before starting: TB test (IGRA or TST), hepatitis B and C serology, vaccines up to date (non-live), and labs (CBC, liver, kidney) every 1–3 months. S57; S57 (notes)

Figure 16 · Biologic DMARDs: mechanism by target. Schematic (illustrative): TNF inhibitors (infliximab, adalimumab, etanercept) neutralize soluble and membrane TNF-α; tocilizumab blocks the IL-6 receptor; abatacept (CTLA-4-Ig) blocks CD28 costimulation of T cells; rituximab binds CD20 and depletes B cells. S52; S52 (notes)

OA vs. RA at a glance

	Osteoarthritis	Rheumatoid arthritis
Disease type	Degenerative	**Autoimmune, systemic inflammatory**
What damages the joint	Cartilage breakdown, mechanical wear	Immune-mediated joint destruction
Treatment goal	Pain control and function	**Suppress disease activity**, not just pain
Core drugs	Acetaminophen, NSAIDs (oral/topical), intra-articular steroids	**DMARDs** (methotrexate first)

How to tell them apart. Sources: S41; S41 (notes); S43; S47

Conventional DMARDs: where each drug acts

Drug	Target / action	Effect
Methotrexate	Inhibits DHFR; disrupts folate metabolism	Slower immune-cell growth
Hydroxychloroquine	Alters lysosomal and TLR signaling	Less antigen signaling

Drug	Target / action	Effect
Sulfasalazine	Reduces cytokine and prostaglandin production	Less inflammation
Leflunomide	Inhibits DHODH; limits pyrimidine synthesis	Slower T-cell growth

How to tell them apart. Sources: S48

Conventional DMARD drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Methotrexate (Trexall, Otrexup)	**Conventional DMARD (antimetabolite, DHFR inhibitor)**	Tablets 2.5 mg; subcutaneous autoinjector; injection	7.5-25 mg ONCE WEEKLY PO or SC, with folic acid 1 mg daily	Rheumatoid arthritis (anchor DMARD), psoriatic arthritis, psoriasis
Hydroxychloroquine (Plaquenil)	**Conventional DMARD (antimalarial)**	Tablets 200 mg	200-400 mg PO daily (maximum 5 mg/kg/day actual body weight)	Rheumatoid arthritis (mild), lupus, malaria
Sulfasalazine (Azulfidine)	**Conventional DMARD**	Tablets 500 mg; DR 500 mg	500 mg PO daily, titrate to 1,000 mg twice daily	Rheumatoid arthritis, ulcerative colitis
Leflunomide (Arava)	**Conventional DMARD (DHODH inhibitor)**	Tablets 10/20 mg	20 mg PO daily (loading 100 mg daily for 3 days optional)	Rheumatoid arthritis

How to tell them apart. Sources: S69; S70; S71; S72

Conventional DMARD drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Methotrexate (Trexall, Otrexup)	Nausea, mucositis, hepatotoxicity, myelosuppression, pneumonitis, teratogenicity	Pregnancy, significant liver disease, severe renal impairment, active infection, heavy alcohol use	Daily dosing by mistake causes severe toxicity; leucovorin rescue; monitor CBC, liver tests and creatinine
Hydroxychloroquine (Plaquenil)	Nausea, rash, headache, retinopathy with long use, QT prolongation	Known retinal disease, hypersensitivity; caution with QT-prolonging drugs	Retinal toxicity with long-term use (baseline and annual eye exams); cardiotoxicity and arrhythmia in overdose
Sulfasalazine (Azulfidine)	Nausea, rash, headache, myelosuppression, hepatotoxicity, orange urine, reversible oligospermia	Sulfa or salicylate allergy, intestinal or urinary obstruction, porphyria; G6PD deficiency (caution)	Agranulocytosis and hemolysis; monitor CBC and liver tests

Drug	Adverse effects	Contraindications	Toxicity / overdose
Leflunomide (Arava)	Diarrhea, hepatotoxicity, hypertension, alopecia, neuropathy, teratogenicity	Pregnancy, severe liver disease, concurrent hepatotoxic drugs	Boxed warning for hepatotoxicity and embryo-fetal toxicity; cholestyramine washout

How to tell them apart. Sources: S69; S70; S71; S72

The four biologic classes in the lecture

Class · example	How it works	Pearl
TNF inhibitors · adalimumab, etanercept, infliximab	Bind soluble or membrane TNF-α, preventing TNFR activation (etanercept is a receptor fusion)	First-line biologic; watch latent TB reactivation
IL-6 inhibitor · tocilizumab	Blocks membrane and soluble IL-6 receptors	Works as monotherapy; can raise lipids and liver enzymes
B-cell therapy · rituximab (anti-CD20)	Binds CD20 and depletes B cells for months	After TNF-inhibitor failure; check hepatitis B first
T-cell costimulation · abatacept (CTLA-4-Ig)	Binds CD80/86; blocks CD28 costimulation	Acts earliest; generally lower infection signal

How to tell them apart. Sources: S51; S52; S53; SG U2-R8

Drug names reveal the class: read the stem

Stem	Meaning	Example
-mab	Monoclonal antibody	Rituximab (anti-CD20)
-zumab	Humanized antibody	Tocilizumab (IL-6R)
-ximab	Chimeric antibody	Infliximab (TNF-α)
-umab	Fully human antibody	Adalimumab (TNF-α)
-cept	Receptor fusion protein	Etanercept, abatacept
-nib / -tinib	Kinase inhibitor	Tofacitinib (JAK)
-coxib	Selective COX-2 inhibitor	Celecoxib
-profen	Propionic-acid NSAID	Ibuprofen, naproxen
-fenac	Acetic-acid NSAID	Diclofenac
-oxicam	Oxicam NSAID	Meloxicam

How to tell them apart. Sources: S60; S60 (notes)

Biologic DMARD drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Adalimumab (Humira)	**Biologic DMARD (TNF inhibitor)**	Prefilled syringe or pen 40 mg/0.8 mL	40 mg SC every other week (weekly for some patients)	Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn disease
Etanercept (Enbrel)	**Biologic DMARD (soluble TNF receptor)**	Prefilled syringe or pen 25/50 mg; vial	50 mg SC once weekly (or 25 mg twice weekly)	Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, plaque psoriasis
Infliximab (Remicade)	**Biologic DMARD (chimeric TNF antibody)**	IV vial 100 mg; biosimilars	3 mg/kg IV at weeks 0, 2 and 6, then every 8 weeks, with methotrexate	Rheumatoid arthritis, Crohn disease, ulcerative colitis, ankylosing spondylitis
Tocilizumab (Actemra)	**Biologic DMARD (IL-6 receptor antagonist)**	IV 20 mg/mL; SC prefilled syringe or pen 162 mg	IV 4-8 mg/kg every 4 weeks; SC 162 mg every other week to weekly	Rheumatoid arthritis, giant cell arteritis, systemic JIA
Abatacept (Orencia)	**Biologic DMARD (T-cell costimulation modulator)**	IV 250 mg vial; SC prefilled syringe 125 mg	IV by weight (500-1,000 mg) at weeks 0, 2, 4 then every 4 weeks; SC 125 mg weekly	Rheumatoid arthritis, psoriatic arthritis, polyarticular JIA
Rituximab (Rituxan)	**Biologic DMARD (anti-CD20 antibody)**	IV 100/500 mg vials	1,000 mg IV on days 1 and 15, repeated no sooner than every 16-24 weeks, with methotrexate	Rheumatoid arthritis (after TNF inhibitor failure), lymphoma, vasculitis

How to tell them apart. Sources: S73; S74; S75; S76; S77; S78

Biologic DMARD drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Adalimumab (Humira)	Injection-site reaction, infection, headache, rash, demyelination (rare), lymphoma (rare)	Active serious infection, untreated latent TB, demyelinating disease, NYHA III-IV heart failure	Boxed warning for serious infections and malignancy; screen for TB and hepatitis B before starting
Etanercept (Enbrel)	Injection-site reaction, infection, headache, demyelination (rare)	Active serious infection, sepsis, untreated latent TB	Boxed warning for serious infections and malignancy; screen for TB and hepatitis B
Infliximab (Remicade)	Infusion reaction, infection, headache, anti-drug antibodies, hepatotoxicity	Active serious infection, untreated latent TB, NYHA III-IV heart failure at high dose	Boxed warning for serious infections and malignancy; infusion reactions treated by slowing or stopping

Drug	Adverse effects	Contraindications	Toxicity / overdose
Tocilizumab (Actemra)	Infection, elevated liver enzymes, neutropenia, hyperlipidemia, GI perforation (rare)	Active serious infection, untreated latent TB; caution with diverticulitis	Boxed warning for serious infections; monitor CBC, liver enzymes and lipids
Abatacept (Orencia)	Headache, infection, nasopharyngitis, infusion reaction	Active serious infection; caution in COPD	Infection risk; do not combine with TNF inhibitors; screen for TB and hepatitis B
Rituximab (Rituxan)	Infusion reaction, infection, hypogammaglobulinemia, hepatitis B reactivation	Active serious infection, severe heart failure; screen hepatitis B first	Boxed warnings for infusion reactions, hepatitis B reactivation and PML; premedicate with acetaminophen, antihistamine and steroid

How to tell them apart. Sources: S73; S74; S75; S76; S77; S78

Supporting drugs at a glance (listed for context; not profiled in detail)

Drug(s)	Why it matters
PPIs and misoprostol	GI protection with long-term NSAIDs
Warfarin and anticoagulants	NSAIDs raise bleeding risk
ACE inhibitors, ARBs and diuretics	NSAIDs add kidney injury (“triple whammy”)
Lithium	NSAIDs raise lithium levels
Systemic corticosteroids	Flare bridge; bone loss and GI risk with NSAIDs
Folic acid and leucovorin	Lower methotrexate toxicity
Isoniazid and rifampin	Latent TB treatment before biologics
Tramadol and opioids	Add-on analgesia for osteoarthritis
Live vaccines	Avoid on biologic or immunosuppressive therapy

How to tell them apart. Sources: S79; S79 (notes)

Conventional synthetic DMARDs (textbook)

Drug	How it works	Main uses	Watch for
Methotrexate	DHFR (slides); AICAR transformylase (textbook)	RA anchor; psoriasis, PsA, JIA	Nausea, stomatitis, liver, marrow, lung; teratogenic

Drug	How it works	Main uses	Watch for
Hydroxychloroquine, chloroquine	Suppress T-cell responses and chemotaxis; dampen type 1 IFN	Mild RA; SLE (lowers mortality); Sjögren	Retinopathy (eye exams); cardiomyopathy; 3–6 months to respond
Sulfasalazine	Split into sulfapyridine + 5-aminosalicylic acid	RA (slows radiologic progression); ulcerative colitis	~30% stop it; rash, toxic epidermal necrolysis, neutropenia; reversible male infertility
Leflunomide	Dihydroorotate dehydrogenase inhibitor	RA (equals methotrexate)	Diarrhea (25%), liver tests, hypertension; half-life 19 days (cholestyramine shortens)
Azathioprine	Becomes 6-thioguanine; suppresses B and T cells	RA, SLE, transplant	Marrow suppression (TPMT, NUDT15), GI upset, cancer, infection
Mycophenolate mofetil	IMPDH inhibitor	SLE (renal and nonrenal), scleroderma lung; not proven in RA	GI, liver, cytopenias, infection; contraindicated in pregnancy
Cyclosporine, tacrolimus, voclosporin	Calcineurin inhibitors (block T-cell cytokines)	Severe RA; lupus nephritis	Nephro-, neuro-, cardiotoxicity, hypertension; CYP3A4 interactions (grapefruit)
Cyclophosphamide	Cross-links DNA	Lupus nephritis, CNS lupus, vasculitis	Cytopenias, infection, sterility, bladder and skin cancer
Apremilast	PDE4 inhibitor (↑ cAMP)	Plaque psoriasis, PsA, Behçet oral ulcers	Diarrhea, nausea, weight loss, mood changes; CYP3A4
Mesalamine	Topical colonic 5-aminosalicylate	Ulcerative colitis	Diarrhea, headache; renal impairment (monitor)

How to tell them apart. Sources: S47; S49; p. 682; p. 683; p. 684; p. 687; p. 688; p. 689; p. 690

Biologic and targeted DMARDs in the reading (textbook)

Drug(s)	Target	Main uses	Watch for
Adalimumab, etanercept, infliximab, certolizumab, golimumab	TNF-α	RA, PsA, AS, psoriasis; infliximab and adalimumab also Crohn and UC	TB, fungal, hepatitis B; heart failure; demyelination; lymphoma (boxed)
Tocilizumab, sarilumab	IL-6 receptor	RA; tocilizumab also giant cell arteritis	Infection, liver enzymes, lipids, cytopenias; no sarilumab with diverticulitis

Drug(s)	Target	Main uses	Watch for
Rituximab	CD20 B cells (depletion)	RA with methotrexate; GPA	First-dose rash (~30%), infusion reactions, hepatitis B reactivation
Abatacept	CD80/86–CD28 costimulation	RA, PsA, JIA	Infection; don't combine with TNF inhibitors; live vaccines
Anakinra, canakinumab, rilonacept	Interleukin-1	CAPS; anakinra approved but rarely used in RA	Injection-site reactions (up to 40%), URIs
Baricitinib, tofacitinib, upadacitinib	Janus kinase (JAK-STAT), oral	RA after TNF or methotrexate failure; PsA; UC	Boxed VTE/MACE/cancer; herpes zoster; ↑ LDL; caution with diverticulitis
Secukinumab, ixekizumab	IL-17A	Psoriasis, PsA, AS	Infection; can cause or worsen IBD; TB screen
Ustekinumab	IL-12/IL-23 (p40)	PsA, psoriasis, Crohn, UC	URIs; stop ≥ 15 weeks before live vaccines
Guselkumab, risankizumab	IL-23 (p19)	Plaque psoriasis, PsA	URIs; TB screen
Belimumab	BLyS (B-cell survival)	SLE, lupus nephritis	Infection; depression and suicide reported
Anifrolumab	Type 1 interferon receptor	Nonrenal SLE	URIs, bronchitis, infusion reactions

How to tell them apart. Sources: S51; p. 682; p. 683; p. 684; p. 685; p. 686; p. 689; p. 690; p. 692; p. 693

SLIDE DISCREPANCY

- **How methotrexate works at RA doses**
- Slides: Inhibits dihydrofolate reductase, reducing immune cell proliferation at low RA doses; the depth slide adds that the low-dose RA effect is likely anti-inflammatory via adenosine release, not cytotoxicity. S49; S50; SG U2-M
- Textbook: Low-dose methotrexate mainly inhibits AICAR transformylase and thymidylate synthetase, raising extracellular AMP that suppresses inflammation; its dihydrofolate reductase effect is relatively minor at rheumatic doses. p. 688

WHY IT HAPPENS

- Why can't RA be treated like OA? RA is immune-mediated joint destruction; analgesics ease pain but leave the immune attack — and the damage — running. S41; S45 (notes)
- Why screen for TB and hepatitis first? These drugs suppress the immune response that keeps latent infections in check, so a hidden infection can reactivate. S56

MNEMONIC (MEMORY AID)

- Methotrexate rules = **W-F-P**: **W**eekly (never daily), **F**olate with it, **P**regnancy contraindicated. S49; SG U2-R7
- Methotrexate toxicity hits **L-B-L**: **L**iver, **B**one marrow, **L**ung. S49
- Generic-name endings: **-mab** = monoclonal antibody (adalimumab, infliximab, tocilizumab, rituximab); **-cept** = receptor-fusion protein (etanercept, abatacept). S60

HIGH-YIELD

- OA = degenerative → pain control; RA = autoimmune → suppress disease activity. S41
- OA ladder: acetaminophen → oral or topical NSAIDs → intra-articular steroids for a single-joint flare. S43
- DMARDs (conventional and biologic — methotrexate, adalimumab, etanercept, infliximab, tocilizumab, rituximab, abatacept) modify RA's disease course rather than only masking symptoms. S61; S45; SG U2-R6
- Methotrexate: DHFR inhibitor, **weekly**, with folate. S49; SG U2-R7
- Methotrexate toxicity: liver, marrow, lung; teratogenic. S49
- HCQ = milder RA; sulfasalazine = combinations; leflunomide = if MTX not tolerated. S47
- DMARDs change disease. Methotrexate is weekly, and screen for TB and hepatitis before biologics. S61 (notes)
- TNF: adalimumab, etanercept, infliximab · IL-6: tocilizumab · B cell: rituximab · T-cell costimulation: abatacept. S51; SG U2-R8
- Shared risk: immunosuppression → infection. S56; SG U2-R8
- Before biologics: TB + hepatitis B/C screening, update (live) vaccines. S56

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Pain by mechanism	Nociceptive · inflammatory · neuropathic S5
Chronic pain	Persists beyond expected healing, often > 3 months S5
4 pain mediators	Prostaglandins, bradykinin, histamine, cytokines (TNF-α, IL-1, IL-6) S6; SG U2-R1
Corticosteroid site	Phospholipase A_2 (upstream) → broad mediator suppression S9; SG U2-R2
NSAID site	COX → prostaglandins specifically S9; S24
Acetaminophen effects	Analgesic + antipyretic; minimal anti-inflammatory S15
Acetaminophen toxic metabolite	NAPQI (N-acetyl-p-benzoquinone imine) S19; S19 (notes)
Acetaminophen antidote	N-acetylcysteine — glutathione precursor S19; SG U2-R3
Hidden acetaminophen	Combination cold, flu, and sleep products — ask about ALL sources S20
COX-1	Housekeeping: gastric protection, renal blood flow S26; SG U2-R4
COX-2	Induced at inflammation; main inflammatory prostaglandin source S26; SG U2-R4
Ibuprofen	Common first-line for mild-moderate pain S28
Naproxen	Longer half-life, twice-daily S28
Diclofenac	Oral and topical S28
Indomethacin	More potent; more GI/CNS effects S28
Ketorolac	Short-term only; significant GI/renal risk S28
Celecoxib trade-off	↓ GI risk, ↑ CV risk, same renal risk S29; SG U2-R5
NSAID adverse effects	GI (ulcer, bleed) · renal (AKI, fluid) · CV (HTN, events) S34
NSAID cautions	GI bleed/ulcer, CKD, HF/CV disease, anticoagulant, late pregnancy S36
Triple whammy	NSAID + ACEi/ARB + diuretic → AKI S36 (notes)
NSAID + corticosteroid	↑ GI ulcer risk S36
OA vs. RA	Degenerative → pain/function vs. autoimmune → suppress disease S41

Prompt	Answer
OA drugs	Acetaminophen, oral/topical NSAIDs, intra-articular steroid for a single-joint flare S43
DMARD	Alters RA's course, not just symptoms S45; SG U2-R6
Conventional DMARDs	Methotrexate (anchor), hydroxychloroquine (milder), sulfasalazine (combos), leflunomide (MTX intolerant) S47
Methotrexate	DHFR inhibitor · weekly · folate · liver/marrow/lung · teratogenic S49; SG U2-R7
TNF inhibitors	Adalimumab, etanercept, infliximab S51; SG U2-R8
IL-6 inhibitor	Tocilizumab S51
B-cell therapy	Rituximab S51
T-cell costimulation	Abatacept S51; SG U2-M
DMARD shared risk	Immunosuppression → infection S56; SG U2-R8
Before biologics	TB + hepatitis B/C screen; update vaccines (live) first S56
Acetaminophen routes	~90% glucuronidation/sulfation (safe) · ~5–10% CYP2E1 → NAPQI S16
Who's at higher acetaminophen risk	Chronic alcohol use, fasting (↓ GSH, ↑ CYP2E1) S17
Liver lesion in overdose	Centrilobular necrosis S17
Rumack-Matthew nomogram	Single acute ingestion; level ≥ 4 h; above the line → NAC; not used > 24 h S18
Nomogram example	8 h, 120 mcg/mL → above the line → start NAC S18
Nomogram line (illustrative)	150 mcg/mL at 4 h, halving every 4 h, 4.7 at 24 h S18
COX-1 vs. COX-2	Constitutive (gastric mucosa, platelet TXA2, kidney) vs. inducible (inflamed tissue) S27
Celecoxib CV mechanism	PGI2 falls, TXA2 preserved S30
NSAID organ effects	GI: ↓ mucus/bicarbonate · kidney: ↓ afferent dilation · platelets: ↓ TXA2 S35
RA joint destruction	Synovitis → pannus; TNF-α and IL-6 drive erosion S42
NSAID vs. DMARD onset	NSAID relief in days · DMARD protection over weeks–months S46
Leflunomide target	DHODH (pyrimidine synthesis) S48
Hydroxychloroquine action	Alters lysosomal and TLR signaling S48
Methotrexate at low dose	Likely anti-inflammatory via adenosine release S50
Abatacept	CTLA-4-Ig: binds CD80/86, blocks CD28 costimulation — acts earliest S52; S55

Prompt	Answer
Rituximab	Anti-CD20; after TNF-inhibitor failure; check hepatitis B S52
Infection monitoring on DMARDs	CBC, liver, kidney labs every 1–3 months S57
Name stems	-zumab humanized · -ximab chimeric · -umab human · -cept fusion · -tinib kinase · -coxib COX-2 S60
Acetaminophen max (adults)	Generally ≤ 4 g/day; 15 g may be fatal p. 694
Aspirin vs. other NSAIDs	Aspirin: irreversible platelet COX (8–10 days); others reversible p. 675; p. 677
Ketorolac limits	Start IV/IM/intranasal; ≤ 120 mg/day; ≤ 5 days; halve if > 65, < 50 kg, renal impairment p. 694
Celecoxib selectivity	10–20× more selective for COX-2 p. 677
Methotrexate dose	7.5 mg weekly → 15–25 mg weekly p. 688
Hydroxychloroquine monitoring	Baseline eye exam; annual after 5 years p. 684
TNF inhibitor boxed warning	Malignancy, especially lymphoma p. 692
JAK inhibitor boxed warning	VTE (DVT/PE), MACE, cancer p. 686
JAK inhibitors	Baricitinib, tofacitinib, upadacitinib — oral; zoster, ↑ LDL p. 686
RA prednisone ceiling	≤ 7.5 mg daily long term, taper p. 693

Self-test

First, the lecture's own 7 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S10-11

In Figure 1, corticosteroids are shown blocking the pathway at the "Membrane Phospholipids → Arachidonic Acid" step, while NSAIDs act one step later. What does this positioning explain?

- **A** Corticosteroids act upstream of NSAIDs, which is why they can suppress a broader range of inflammatory mediators, not just prostaglandins
- **B** Corticosteroids and NSAIDs block the exact same enzyme
- **C** NSAIDs act upstream of corticosteroids
- **D** Neither drug class affects the arachidonic acid pathway

ANSWER: A

By inhibiting phospholipase A_2, corticosteroids block the release of arachidonic acid itself — upstream of COX — which is why they suppress a broader range of inflammatory mediators (not just prostaglandins) than NSAIDs, which act only at the COX step. S10-11

Lecture check 2 · S12-13

Which mediator is the primary target of NSAID therapy within the inflammatory pain pathway?

- **A** Prostaglandins
- **B** Histamine
- **C** Bradykinin
- **D** Cytokines broadly (e.g., all of TNF-α, IL-1, IL-6)

ANSWER: A

NSAIDs inhibit COX enzymes, directly reducing prostaglandin synthesis. Histamine, bradykinin, and cytokines are important inflammatory mediators too, but they are not the primary target of NSAID mechanism. S12-13

Lecture check 3 · S21-22

In acetaminophen overdose, hepatotoxicity results primarily from which mechanism?

- **A** Direct COX inhibition in the liver
- **B** Excess Phase II glucuronidation
- **C** Accumulation of NAPQI after glutathione stores are depleted
- **D** Renal crystal formation

ANSWER: C

NAPQI, normally detoxified by conjugation with glutathione, accumulates once glutathione stores are exhausted in overdose, causing hepatocyte injury — which is why N-acetylcysteine (a glutathione precursor) is the antidote. S21-22

Lecture check 4 · S32-33

Based on Figure 2, a patient with a history of GI ulcer but no cardiovascular disease is switched from ibuprofen to celecoxib. What is the most likely net effect on this patient's risk profile?

- **A** Lower GI risk, with some increase in cardiovascular risk
- **B** Both GI and cardiovascular risk decrease
- **C** No change in either risk, since all NSAIDs carry identical risk profiles
- **D** Both GI and cardiovascular risk increase

ANSWER: A

Figure 2 shows celecoxib's illustrative GI-risk bar is lower than the nonselective NSAID bar, while its cardiovascular-risk bar is higher — exactly the trade-off described in the celecoxib bullet slide. For this GI-ulcer-history patient without cardiovascular disease, that trade may be favorable, but the increased cardiovascular risk should still be discussed. S32-33

Lecture check 5 · S38-39

A patient with CKD, hypertension, and anticoagulant use needs analgesia for osteoarthritis. Which class carries the highest combined renal and bleeding risk for this patient?

- **A** Acetaminophen
- **B** Oral NSAIDs
- **C** Topical capsaicin
- **D** Physical therapy alone

ANSWER: B

This patient has multiple NSAID red flags at once: CKD (renal risk), anticoagulant use (bleeding risk), and hypertension (cardiovascular risk) — oral NSAIDs carry the highest combined risk here. S38-39

Lecture check 6 · S54-55

In Figure 3, which biologic DMARD acts earliest in the cascade shown, before cytokines like TNF-α and IL-6 are even released?

- **A** Adalimumab
- **B** Tocilizumab
- **C** Abatacept
- **D** Rituximab

ANSWER: C

Abatacept blocks T-cell costimulation — the step furthest upstream in the diagram, occurring before activated T cells drive cytokine release (blocked downstream by adalimumab/tocilizumab) or B-cell activity (blocked by rituximab). S54-55

Lecture check 7 · S58-59

Which feature distinguishes a DMARD from an NSAID in the management of rheumatoid arthritis?

- **A** DMARDs provide faster pain relief than NSAIDs
- **B** NSAIDs suppress the immune system more potently than DMARDs
- **C** DMARDs modify the underlying disease process rather than only masking symptoms
- **D** DMARDs are only available as biologic agents

ANSWER: C

An NSAID can make a patient feel better without changing RA's disease trajectory; a DMARD (conventional or biologic) is intended to slow or halt joint destruction and systemic inflammation itself. S58-59

Original questions

Q1 · Mechanism · Acetaminophen

In acetaminophen overdose, why is N-acetylcysteine effective?

- **A** It speeds renal excretion of unchanged acetaminophen
- **B** It is a glutathione precursor that restores NAPQI detoxification
- **C** It blocks COX in the liver and lowers prostaglandin levels
- **D** It inhibits the enzyme that forms NAPQI from the drug
- **E** It binds acetaminophen in the gut and prevents absorption

ANSWER: B

Overdose depletes glutathione so NAPQI accumulates; N-acetylcysteine replenishes glutathione to detoxify NAPQI (S19; SG U2-R3). S19; SG U2-R3

- **A.** The antidote works by refilling glutathione, not by renal excretion (S19).
- **C.** COX blockade is the NSAID mechanism, not the antidote's (S24).
- **D.** The lecture's mechanism is glutathione replenishment (SG U2-R3).
- **E.** The lecture explains the antidote through glutathione, not binding (S19).

Q2 · Clinical vignette · Acetaminophen

A 40-year-old with an acutely swollen, inflamed knee asks for the drug that best treats the inflammation itself. Why is acetaminophen not the best choice?

- **A** It only works when given intravenously in hospital
- **B** It causes more GI bleeding than the NSAIDs do
- **C** It is contraindicated in anyone with joint swelling
- **D** Its anti-inflammatory effect at the periphery is minimal
- **E** It has no analgesic effect in musculoskeletal pain

ANSWER: D

Acetaminophen relieves pain and fever centrally, but its peripheral anti-inflammatory effect is minimal — it is not a reliable anti-inflammatory agent (S15). S15

- **A.** No route restriction is taught; it is used orally first-line for mild OA pain (S43).
- **B.** GI bleeding is the NSAID risk (S34).
- **C.** No such contraindication is taught; the issue is weak anti-inflammatory action (S15).
- **E.** It does relieve mild-to-moderate pain (S15).

Q3 · Mechanism · NSAIDs & COX-2 Selective Inhibition

Which statement best explains why nonselective NSAIDs cause GI and renal adverse effects?

- **A** They block TNF-α, which normally protects the stomach lining
- **B** They block phospholipase A_2 and stop all mediator release
- **C** They inhibit COX-1, which protects gastric mucosa and renal blood flow
- **D** They inhibit COX-2, the only source of protective prostaglandins
- **E** They deplete glutathione, which injures the gut and kidney

ANSWER: C

COX-1 is the housekeeping enzyme that maintains gastric mucosal protection and normal renal blood flow; inhibiting it contributes to GI and renal adverse effects (S26; SG U2-R4). S26; SG U2-R4

- **A.** TNF-α blockade is a biologic DMARD mechanism (S51).
- **B.** That is the corticosteroid site (S9).
- **D.** COX-2 is the main source of inflammatory prostaglandins (S26).
- **E.** Glutathione depletion is acetaminophen overdose (S19).

Q4 · Clinical vignette · NSAIDs & COX-2 Selective Inhibition

A patient on lisinopril (an ACE inhibitor) and hydrochlorothiazide starts daily naproxen for knee pain. Two weeks later, creatinine has risen sharply. What pattern explains this?

- **A** Expected COX-2 renal protection failing under stress
- **B** Bleeding from the NSAID plus the ACE inhibitor together
- **C** The “triple whammy”: NSAID + ACE inhibitor + diuretic causing AKI
- **D** A GI ulcer from NSAID plus diuretic, causing dehydration
- **E** Hepatotoxicity from NAPQI formed by naproxen metabolism

ANSWER: C

NSAIDs with ACE inhibitors/ARBs raise AKI risk, and adding a diuretic is the high-yield “triple whammy” AKI pattern (S36; S36 notes). S36; S36 (notes)

- **A.** COX-2 selectivity does not protect the kidney, and naproxen is nonselective (S28-29).
- **B.** NSAID + anticoagulant raises bleeding; this is a renal pattern (S36).
- **D.** NSAID + corticosteroid raises ulcer risk; this is a kidney injury pattern (S36).
- **E.** NAPQI comes from acetaminophen (S19).

Q5 · Mechanism · NSAIDs & COX-2 Selective Inhibition

Which nonselective NSAID does the lecture restrict to short-term use because of significant GI and renal risk?

- **A** Ketorolac
- **B** Ibuprofen
- **C** Diclofenac
- **D** Celecoxib
- **E** Naproxen

ANSWER: A

Ketorolac is for short-term use only because of significant GI/renal risk (S28). S28

- **B.** Ibuprofen is a common first-line choice for mild-moderate pain (S28).
- **C.** Diclofenac is noted for oral and topical formulations (S28).
- **D.** Celecoxib is COX-2 selective; its trade-off is cardiovascular risk (S29).
- **E.** Naproxen is noted for its longer half-life and twice-daily dosing (S28).

Q6 · Clinical vignette · NSAIDs & COX-2 Selective Inhibition

A patient takes low-dose aspirin for cardioprotection and starts regular ibuprofen for back pain. What is the textbook concern?

- **A** Ibuprofen converts aspirin to the toxic metabolite NAPQI
- **B** Ibuprofen speeds aspirin's renal excretion fivefold
- **C** Ibuprofen blocks aspirin's irreversible platelet inhibition
- **D** Aspirin makes ibuprofen COX-2 selective, raising CV risk
- **E** Both drugs together protect the kidney from injury

ANSWER: C

Concomitant ibuprofen antagonizes aspirin's irreversible platelet inhibition, limiting its cardioprotective effect (p. 680). p. 680

- **A.** NAPQI comes from acetaminophen (S19).
- **B.** Not the textbook's interaction (p. 680).
- **D.** Aspirin does not change ibuprofen's selectivity (p. 680).
- **E.** NSAIDs reduce renal perfusion; they don't protect the kidney (S34).

Q7 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A 66-year-old has knee pain worse with activity, no systemic symptoms, and mild disease. Which first-line drug does the lecture call reasonable?

- **A** Acetaminophen for mild osteoarthritis pain
- **B** Methotrexate to slow joint destruction
- **C** Hydroxychloroquine for mild disease
- **D** Adalimumab to block TNF-α signaling
- **E** Ketorolac taken daily as maintenance

ANSWER: A

Acetaminophen is a reasonable first-line for mild OA pain (S43). S41; S43

- **B.** Methotrexate is the anchor DMARD for RA, an autoimmune disease (S41; S47).
- **C.** Hydroxychloroquine is a DMARD for milder RA (S47).
- **D.** Biologic DMARDs treat RA, not degenerative OA (S41; S51).
- **E.** Ketorolac is short-term only (S28).

Q8 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

What is the key difference in treatment goals between osteoarthritis and rheumatoid arthritis?

- **A** Both aim only at pain relief with NSAIDs
- **B** OA aims at cartilage regrowth with DMARDs
- **C** OA aims at pain and function; RA aims to suppress disease activity
- **D** OA aims to suppress immunity; RA aims at pain relief only
- **E** RA aims at weight loss; OA aims at infection control

ANSWER: C

OA treatment targets pain control and function; RA treatment must suppress disease activity, not just pain (S41). S41; S41 (notes)

- **A.** RA needs disease-modifying therapy (S41 notes).
- **B.** OA is managed mainly with analgesics/NSAIDs (S41 notes).
- **D.** Reversed (S41).
- **E.** Neither is the taught goal (S41-43).

Q9 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A patient newly started on methotrexate for RA asks whether to take a tablet every morning "to keep levels steady." What is the correct counseling?

- **A** Take it daily but skip the folate supplement
- **B** Take it only on days when joints are painful
- **C** Take it twice daily, like naproxen, for steady levels
- **D** Take it once weekly, because daily dosing can be fatal
- **E** Take it daily with food to reduce stomach upset

ANSWER: D

Methotrexate is dosed weekly, not daily — a critical patient-safety point, since daily dosing can be fatal (S49; SG U2-R7). S49; SG U2-R7

- **A.** Folate is co-prescribed to reduce toxicity, and dosing is weekly (S49).
- **B.** It is a scheduled weekly DMARD, not an as-needed analgesic (S45; S49).
- **C.** Twice-daily dosing describes naproxen (S28).
- **E.** Daily dosing is the dangerous error (S49).

Q10 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

What is methotrexate's mechanism as taught in the lecture, and what is co-prescribed to reduce its toxicity?

- **A** COX-2 inhibition; a proton pump inhibitor
- **B** Phospholipase A_2 blockade; calcium supplementation
- **C** TNF-α blockade; tuberculosis screening
- **D** Glutathione depletion; N-acetylcysteine
- **E** Dihydrofolate reductase inhibition; folate supplementation

ANSWER: E

Methotrexate inhibits dihydrofolate reductase, reducing immune cell proliferation at low RA doses; folate is routinely co-prescribed (S49; SG U2-R7; SG U2-M). S49; SG U2-M; SG U2-R7

- **A.** COX-2 inhibition is celecoxib (S29).
- **B.** Phospholipase A_2 is the corticosteroid site (S9).
- **C.** TNF blockade is the biologic DMARD mechanism (S51).
- **D.** That is acetaminophen toxicity and its antidote (S19).

Q11 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A 29-year-old with active RA who is planning a pregnancy needs a DMARD. Which drug from the lecture is contraindicated?

- **A** Celecoxib, because it is a disease-modifying drug
- **B** Sulfasalazine, because it is only used alone
- **C** Acetaminophen, because it is a DMARD
- **D** Methotrexate, because it is teratogenic
- **E** Hydroxychloroquine, because of its poor safety profile

ANSWER: D

Methotrexate is contraindicated in pregnancy because it is teratogenic (S49). S49

- **A.** Celecoxib is an NSAID, not a DMARD (S29; S45).
- **B.** Sulfasalazine is often used in combination regimens (S47).
- **C.** Acetaminophen is an analgesic, not a DMARD (S15; S45).
- **E.** Hydroxychloroquine has a relatively favorable safety profile (S47).

Q12 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Before starting adalimumab for rheumatoid arthritis, which baseline step does the lecture specifically call for?

- **A** Screening for tuberculosis and hepatitis B/C
- **B** Starting folate to prevent bone marrow toxicity
- **C** Adding an ACE inhibitor to protect the kidney
- **D** Checking for all hidden acetaminophen sources
- **E** Avoiding all vaccines until therapy has begun

ANSWER: A

Biologics cause immunosuppression; baseline tuberculosis and hepatitis B/C screening comes before starting them (S56). S56

- **B.** Folate is co-prescribed with methotrexate (S49).
- **C.** Not a DMARD safety step; NSAID + ACEi raises AKI risk (S36).
- **D.** That is acetaminophen safety counseling (S20).
- **E.** Vaccines, especially live vaccines, are updated before therapy (S56).

Q13 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A patient on a biologic DMARD calls with a fever of 38.6 °C and a productive cough. What lecture principle drives the advice to be seen promptly?

- **A** NSAID-like renal toxicity makes fever dangerous
- **B** Fever means the biologic is working as intended
- **C** Immunosuppression raises infection risk across all DMARDs
- **D** Biologics raise cardiovascular risk like celecoxib
- **E** Biologics cause hidden acetaminophen overdose

ANSWER: C

The major safety theme across all DMARDs is immunosuppression → increased infection risk; patients should report fevers or infection symptoms promptly (S56; SG U2-R8). S56; SG U2-R8

- **A.** Renal toxicity is an NSAID theme, not the DMARD safety theme (S34; S56).
- **B.** Fever signals possible infection in an immunosuppressed patient (S56).
- **D.** That trade-off belongs to celecoxib (S29).
- **E.** Hidden acetaminophen comes from combination cold products (S20).

Q14 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Match the drug to its target: which pairing is correct?

- **A** Abatacept — B-lymphocyte depletion
- **B** Tocilizumab — IL-6 receptor signaling
- **C** Etanercept — IL-6 receptor signaling
- **D** Adalimumab — T-cell costimulation
- **E** Rituximab — TNF-α signaling

ANSWER: B

Tocilizumab is the lecture's IL-6 inhibitor (S51; SG U2-R8). S51; SG U2-R8

- **A.** Abatacept modulates T-cell costimulation (S51).
- **C.** Etanercept blocks TNF-α (S51).
- **D.** Adalimumab blocks TNF-α (S51).
- **E.** Rituximab depletes B cells; TNF is adalimumab, etanercept, infliximab (S51).

Q15 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A patient with RA and high cardiovascular risk is being considered for tofacitinib. Which boxed warning applies to all JAK inhibitors?

- **A** Lymphoma only, exactly the same as the TNF inhibitors
- **B** Teratogenicity that requires strict weekly-only dosing
- **C** Retinopathy that requires annual eye examinations
- **D** Venous thromboembolism, major cardiovascular events, and cancer
- **E** Fatal hepatotoxicity from NAPQI accumulating in the liver

ANSWER: D

All JAK inhibitors carry an FDA boxed warning for VTE (DVT, PE), MACE, and cancer (p. 686). p. 686

- **A.** The TNF inhibitor warning is malignancy; JAK inhibitors add VTE and MACE (p. 686; p. 692).
- **B.** Weekly dosing is a methotrexate rule (S49).
- **C.** Retinal screening applies to hydroxychloroquine (p. 684).
- **E.** NAPQI is acetaminophen toxicity (S19).

Q16 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Which conventional DMARD requires a baseline eye exam and annual screening after 5 years because of retinopathy?

- **A** Methotrexate
- **B** Mycophenolate mofetil
- **C** Sulfasalazine
- **D** Leflunomide
- **E** Hydroxychloroquine

ANSWER: E

Antimalarials bind melanin-rich tissue such as the retina; a baseline exam and annual screening after 5 years are recommended (p. 684). p. 684

- **A.** Methotrexate monitoring targets liver, marrow, and lung (S49).
- **B.** Mycophenolate's issues are GI, liver, cytopenias, and infection (p. 689).
- **C.** Sulfasalazine's issues are GI, rash, and cytopenias (p. 690).
- **D.** Leflunomide's issues are diarrhea and liver tests (p. 688).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO2.1** Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.
- [ ] **IO2.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.
- [ ] **IO2.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO2.4** Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-specific factors.
- [ ] **IO2.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.
- [ ] **IO2.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal.

- [ ] **SO1** Explain the pharmacologic basis of musculoskeletal pain and inflammation.
- [ ] **SO2** Compare acetaminophen and NSAIDs, including COX-1 vs. COX-2 inhibition.
- [ ] **SO3** Identify major NSAID adverse effects and select therapy based on patient risk factors.
- [ ] **SO4** Explain the mechanisms of conventional vs. biologic DMARDs, including representative generic agents in each class.
- [ ] **SO5** Identify key contraindications, adverse effects, and drug interactions for analgesics and DMARDs.
- [ ] **SO6** Preview: gout, skeletal muscle relaxants, and bone pharmacology continue in Session 2 (Thursday, October 29).